

RICHMOND OPERATION CENTRE, BC, Canada

WMO: 711120

Lat: 49.182N

Lon: 123.078W

Elev: 52

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.1	32.1	14.4	11.4	33.6	20.9	15.7	34.5	38.1	46.0	32.7	44.3	4.7	70	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	66.0	59.3	63.5	58.6	61.9	57.8	60.3	64.1	59.3	62.4	58.4	61.1	7.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.5	73.4	61.6	57.5	70.8	60.5	57.0	69.3	59.8	26.6	64.6	25.8	62.5	25.3	61.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.0	26.8	22.8	25.7	73.2	4.8	2.0	22.2	74.6	19.4	75.8	16.7	76.9	13.2	78.4		
(5)			22.5	63.4	5.0	2.2	18.8	65.0	15.9	66.2	13.0	67.5	9.4	69.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	42.5	42.7	44.6	47.3	51.1	54.2	56.7	57.6	55.8	50.9	45.8	42.6		
	DBStd	6.58	4.24	5.31	3.43	3.17	3.17	2.73	2.42	2.32	3.13	3.12	3.77	4.61		
	HDD50	1115	234	207	170	92	22	1	0	0	1	26	132	230		
	HDD65	5716	698	625	633	532	432	324	258	231	276	437	576	694		
	CDD50	876	1	2	1	11	55	128	207	235	175	54	6	1		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	9.2	11.3	9.8	10.0	9.6	9.3	9.3	7.6	6.9	6.3	7.9	11.0	11.0		
(15) Precipitation	PrecAvg	47.6	6.3	5.0	4.4	3.0	2.6	1.9	1.5	1.6	2.5	4.8	6.9	7.3		
	PrecMax	59.8	10.8	11.3	7.8	5.7	5.8	5.4	3.4	4.2	5.4	11.6	14.1	12.3		
	PrecMin	33.2	1.1	1.3	1.9	0.6	0.9	0.6	0.0	0.0	0.0	0.8	1.9	2.2		
	PrecStd	7.3	2.4	2.4	1.3	1.3	1.2	1.2	1.1	1.1	1.4	2.5	2.6	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	54.6	55.1	60.1	63.4	69.0	69.2	68.4	69.2	61.7	54.7	52.3		
		MCWB	47.1	49.0	48.3	50.6	56.5	57.4	61.1	61.4	59.8	55.8	51.8	48.7		
	2%	DB	49.9	52.0	52.5	56.5	60.4	62.3	63.8	64.9	65.6	58.7	53.6	50.4		
		MCWB	48.0	47.5	47.8	50.1	55.1	57.3	59.3	60.0	58.5	54.1	52.0	49.0		
	5%	DB	48.6	50.1	51.0	54.0	57.7	60.0	62.0	62.9	62.7	56.8	52.2	49.8		
		MCWB	47.2	47.5	47.3	49.7	54.0	56.0	58.2	59.1	57.9	53.5	50.9	48.8		
	10%	DB	48.0	48.6	49.8	52.2	56.1	58.4	60.7	61.5	60.5	55.3	51.4	48.5		
		MCWB	46.7	46.9	47.0	48.9	52.6	55.1	57.3	58.3	57.1	53.0	50.2	47.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	50.9	50.9	53.2	58.1	59.3	61.5	62.0	61.2	57.0	53.7	50.6		
		MCDB	50.2	52.3	51.9	55.8	61.6	64.2	67.2	66.7	66.3	59.5	53.8	51.0		
	2%	WB	48.8	49.4	49.8	51.8	56.1	57.7	59.5	60.5	59.4	55.5	52.6	49.8		
		MCDB	48.9	50.2	50.8	54.6	59.3	61.2	63.2	63.8	63.1	57.2	53.0	50.0		
	5%	WB	47.9	48.4	48.6	50.6	54.6	56.5	58.5	59.5	58.4	54.6	51.7	48.8		
		MCDB	48.1	49.1	49.8	53.0	57.3	59.3	61.4	62.3	61.6	56.0	52.4	49.1		
	10%	WB	46.8	47.4	47.7	49.7	53.1	55.5	57.5	58.6	57.3	53.7	50.4	47.8		
		MCDB	47.3	48.3	49.1	51.7	55.5	57.8	60.1	60.9	60.1	54.9	51.1	48.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.5	7.5	7.6	7.7	6.7	6.8	6.5	7.0	6.5	6.5	6.1		
		MCDBR	5.8	8.2	9.6	10.3	9.7	9.2	8.9	9.1	10.5	8.6	6.1	5.5		
	5% WB	MCWBR	5.2	5.6	7.3	7.2	6.6	6.3	5.6	5.9	6.4	6.3	6.0	5.1		
		MCWBR	4.8	6.2	6.6	8.0	8.8	8.0	8.4	8.2	9.1	6.8	5.5	4.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.320	0.314	0.331	0.364	0.390	0.364	0.362	0.365	0.349	0.338	0.322	0.309		
		taud	2.437	2.500	2.474	2.388	2.347	2.442	2.474	2.478	2.526	2.539	2.498	2.450		
	Ebn at Noon	Ebn at Noon	233	268	281	280	276	284	283	277	272	255	232	221		
		Edh at Noon	20	24	29	35	38	35	33	31	27	23	19	18		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	272	545	819	1274	1666	1789	1998	1695	1175	645	321	218		
	RadStd	RadStd	44	82	112	137	144	212	175	132	105	89	60	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page